

Cloud Computing & DevOps – AWS Handbook

SESSION 2

Amazon EC2 & Web Application Hosting

Deploying Your First Website on AWS Cloud Infrastructure

Duration	2 Hours
Prerequisites	AWS Account, Basic Linux Knowledge
Learning Outcome	Launch EC2, Host Website, Access via Browser

Why Are We Learning This?

Cloud applications require computing infrastructure. Amazon EC2 provides on-demand virtual machines that can host websites, applications and services.

Architecture Diagram



Key Concepts

Server	Computer providing services
Virtual Machine	Software-based computer
EC2	AWS Virtual Machine service
Public IP	Internet-accessible address
Security Group	Virtual firewall

Practical Activities

Activity 1: Launch an EC2 Instance

Expected: Instance State = Running

Activity 2: Configure Security Group

Expected: SSH and HTTP Allowed

Activity 3: Connect to Linux Server

Expected: Successful Login

Activity 4: Install Web Server

Expected: Service Running

Activity 5: Deploy Website

Expected: Website Accessible

What You Should See

Paste Screenshot Here

EC2 Dashboard

Paste Screenshot Here

Instance Details

Paste Screenshot Here

Security Group Rules

Paste Screenshot Here

Linux Login Screen

Paste Screenshot Here

Website in Browser

Journal Writing Template

Aim

To deploy and host a website using Amazon EC2.

Theory

Amazon EC2 provides scalable virtual servers in the cloud.

Procedure

Write 5–7 concise steps performed during the lab.

Observations

EC2 launched successfully. Security Group configured. Website accessible.

Conclusion

Amazon EC2 was successfully used to host a website.

Viva Preparation

1. What is EC2?
2. What is a Virtual Machine?
3. What is a Public IP?
4. What is a Security Group?
5. Difference between Static and Dynamic Websites?

Troubleshooting Box

Problem	Possible Cause
Website not opening	HTTP blocked
Connection failure	Security Group issue
Server inaccessible	Instance stopped

Industry Relevance

Most cloud-hosted applications ultimately run on virtual machines. Understanding EC2 is essential before learning Docker, Kubernetes and DevOps automation.